

Smaller V_{DD} : $\overline{F}_{\text{OR}} > 0.99$ via strong blockade and sub- $\pi/4$ area

Rb $54D_{3/2}$ + Cs $62D_{5/2}$ at $a = 5.00 \mu\text{m}$, smooth super-Gaussian probe with $\alpha = 4$, area = $0.95 \pi/4$

TriQG examples/Smaller_VDD_energy · compiled 2026-05-14 (rev. 4)

Headline results. All on the new energy levels (Rb $54D_{3/2}$ + Cs $62D_{5/2}$, $V_{ct} = 2\pi \times 226 \text{ MHz}$):

- **OR gate:** $\overline{F}_{\text{OR}} = \mathbf{0.992422}$, infidelity $1 - \overline{F}_{\text{OR}} = \mathbf{7.58 \times 10^{-3}}$. Total gate time 385 ns.
- **CCX gate:** $\overline{F}_{\text{CCX}} = \mathbf{0.995854}$, infidelity $1 - \overline{F}_{\text{CCX}} = \mathbf{4.15 \times 10^{-3}}$. Total gate time 95 ns.
- **Both gates above 0.99.** X-round per-cycle infidelity $1 - \overline{F}_{\text{OR}}\overline{F}_{\text{CCX}} \approx 1.17 \times 10^{-2}$.

The OR-gate improvement (from 0.97 in rev. 2 to 0.992 in this revision) comes from **two** changes from rev. 2: (i) restore the paper’s amplitudes $\Omega_p = \Omega_c = 2\pi \times 50 \text{ MHz}$, $\Omega_R = 3.5 \Omega_p$ (instead of the rev. 2 value of $2\pi \times 20 \text{ MHz}$, which was too small to saturate the blockade in a short gate time); (ii) drop the two-photon area from the paper’s $\pi/4$ to $0.95 \pi/4$ – empirically the protocol’s correct target rotation at finite V_{ct} .

Rev. 4 audit. A full $(\Delta, \Omega_R/\Omega_p)$ sweep with ratio relaxed from the Farouk 3.5 confirms that this configuration sits on the **broadest** $\overline{F}_{\text{OR}} > 0.99$ plateau in the (Δ, Ω_R, K) landscape; a secondary peak at $(\Delta = 225 \text{ MHz}, \Omega_R/\Omega_p = 2.5)$ matches the fidelity in $2.1 \times$ shorter gate time but is a knife-edge (Sec. 5).

1 Motivation

The paper’s X-round Rydberg pair (Rb $66D_{5/2}$ + Cs $76D_{3/2}$) sits $\sim 160 \times$ above a Rb same-species Förster zero crossing, so the data–data blockade at $r_{\text{DD}} = a = 5 \mu\text{m}$ reaches

$$V_{\text{DD}}/(2\pi) \approx 91 \text{ MHz},$$

one-sixth of the desired Rb–Cs interspecies blockade. The species-selective global drive in the X-round excites every Rb data atom in parallel, so this V_{DD} enters the per-gate selectivity budget on equal footing with the published V_{cc} entry:

$$R_{\text{DD}} = \frac{V_{ct}}{V_{\text{DD}}} \approx 5.7 \rightarrow R_{\text{DD}} \ll 100 \text{ (target)}.$$

Moving Rb to $54D_{3/2}$ parks the data atom **on** the Rb $D_{3/2}$ same-species zero crossing, killing $|C_6^{\text{DD}}|$ by $\sim 157 \times$. Cs $62D_{5/2}$ keeps the interspecies Förster channel near-resonant ($\Delta_F/(2\pi) = -5.2 \text{ MHz}$). All numerical inputs are imported from `level_parameters.py`; see `SelfCorrectingRydberg/plan/Rb-Cs_level_redesign.md` for the full ARC scan.

The redesigned V_{ct} is $\sim 2.3 \times$ smaller than the paper’s value, which makes the protocol’s $V_{ct} \rightarrow \infty$ assumptions slightly off. The pulse-engineering work in this report fixes that.

2 Setup

2.1 Rydberg levels and Förster channel

- Rb data $|R\rangle = |54D_{3/2}\rangle$, Cs ancilla $|r\rangle = |62D_{5/2}\rangle$, Rb intermediate $|P\rangle = |7P_{3/2}\rangle$ (unchanged).
- Förster channel $|54D_{3/2}; 62D_{5/2}\rangle \leftrightarrow |55P_{3/2}; 60F_{5/2}\rangle$, defect $\Delta_F/(2\pi) = -5.2 \text{ MHz}$.

2.2 Lattice geometry (unchanged)

- Rotated-lattice spacing $a = 5.00 \mu\text{m}$.

- Data-ancilla $r_{\text{DA}} = a/\sqrt{2} \approx 3.5355 \mu\text{m}$.
- Data-data $r_{\text{DD}} = a = 5.0000 \mu\text{m}$.
- Ancilla-ancilla $r_{\text{AA}} = a\sqrt{2} \approx 7.0711 \mu\text{m}$.

2.3 Interaction strengths

Quantity	Paper (Option A)	This redesign
Rydberg pair	Rb $66D_{\frac{5}{2}}$ + Cs $76D_{\frac{3}{2}}$	Rb $54D_{\frac{3}{2}}$ + Cs $62D_{\frac{5}{2}}$
Förster channel	$67P_{\frac{3}{2}} - 74F_{\frac{5}{2}}$	$55P_{\frac{3}{2}} - 60F_{\frac{5}{2}}$
$\Delta_F/(2\pi)$ (MHz)	+10.9	−5.2
$\tilde{C}_3$ (GHz· μm^3)	22.84	10.0
$ C_6^{\text{RbRb}} $ (GHz· μm^6)	~ 1420	9.07
C_6^{CsCs} (GHz· μm^6)	−692.9	−91.0
$V_{ct}/(2\pi)$ (MHz)	+516.81	+226.27
$V_{\text{DD}}/(2\pi)$ (MHz)	+91	+0.58
$V_{cc}/(2\pi)$ (MHz)	−5.54	−0.7280
$R_{\text{DD}} = V_{ct}/V_{\text{DD}}$	~ 5.7 (fail)	389.8 ✓
$R_{\text{AA}} = V_{ct}/ V_{cc} $	~ 94	310.8 ✓

Table 1: Rydberg interaction strengths at $a = 5.00 \mu\text{m}$. The new selectivity ratios R_{DD} and R_{AA} both clear the ≥ 100 threshold.

2.4 Pulse-sequence parameters (FINAL)

Parameter	Paper	Rev. 2	FINAL (rev. 3)
$\Omega_c/(2\pi)$ (MHz)	50	20	50
$\Omega_p/(2\pi)$ (MHz)	50×1.040	20	50
Ω_R/Ω_p	3.5	3.5	3.5
$\Delta/(2\pi)$ (MHz)	500	500	500
T_c (ns)	10	25	10
Shape factor $\alpha = T_f^3/\sigma$	2.41 (abrupt)	4.0 (smooth)	4.0 (smooth)
Edge / peak ratio	3×10^{-3}	1.1×10^{-7}	1.1×10^{-7}
Area / ($\pi/4$)	1.000	1.000	0.950
T_f (ns)	150	1200	182.46
σ (ns)	1.4	432	1.519
Total OR gate (ns)	320	2441	384.9
$\Omega_{cc}/(2\pi)$ (MHz)	100	50	50
$\Omega_t/(2\pi)$ (MHz)	50	20	20
Total CCX gate (ns)	40	95	95

Table 2: Pulse parameters used in this report. The OR-gate target probe is a super-Gaussian of order 6, $\Omega_{p(t)} = (\Omega_p/2) \exp\left[-\left((t - t_c)^3/\sigma\right)^2\right]$ with $t_c = T_c + T_f$, active on $[T_c, T_c + 2T_f]$. All quantities marked **bold** are this report’s settings; the “Rev. 2” column shows the prior overly-conservative configuration.

2.5 Decoherence rates (T = 300 K, BBR-included)

Cs $|62D_{\frac{5}{2}}\rangle$: $\tau_r \approx 77 \mu\text{s}$ ¹ Rb $|54D_{\frac{3}{2}}\rangle$: $\tau_R \approx 74 \mu\text{s}$ ² Rb $|7P_{\frac{3}{2}}\rangle$: $\tau_P = 0.131 \mu\text{s}$ (paper).

¹ n^3 -scaled estimate from ARC values for paper levels; re-verify with `Cs.getStateLifetime(62, 2, 2.5, temperature=300, includeLevelsUpTo=92)`.

3 Strong-blockade audit

For the OR-gate Raman drive on the target, the relevant off-resonant processes have effective rates that must lie **well below** the Rydberg blockade V_{ct} :

Process	Expression	Value / (2π) (MHz)	Margin $V_{ct}/(\cdot)$
One-photon AC Stark on $ P\rangle$	$\Omega_p^2/(2\Delta)$	2.50	90
Two-photon Raman on $ R\rangle$	$\Omega_p\Omega_R/(2\Delta)$	8.75	25.9
Rydberg blockade $V_{ct}/(2\pi)$	$\tilde{C}_3/r_{DA}^3$	226.27	—

Table 3: Strong-blockade conditions at the final configuration. Both off-resonant target rates are at least $26 \times$ below the interspecies blockade; the AC-Stark rate is $90 \times$ below.

Both margins comfortably satisfy “ $\ll V_{ct}$ ” in the sense the user requested; the binding constraint is the two-photon Raman rate (factor **26** below V_{ct}). Tightening these margins further (e.g. by raising Δ) lengthens the gate quadratically and **worsens** fidelity through Cs $|r\rangle$ decay during the gate window; the optimum we converged on sits on the boundary between the blockade-leakage regime (small Δ) and the decay-during-dwell regime (large Δ).

4 Pulse shape and the area = $0.95 \pi/4$ optimum

4.1 Why the area is not $\pi/4$ at finite V_{ct}

The Farouk-style OR protocol was designed in the $V_{ct} \rightarrow \infty$ limit, where the blockade is **perfect**: any control in $|r\rangle$ fully shifts the target’s $|R\rangle$ off resonance. Under that ideal, the two-photon Raman area must equal $\pi/4$ for the correct OR truth table.

At finite V_{ct} , the protocol has small residual leakage on the blocked branches. Empirically, the protocol’s correct two-photon area shifts to compensate. A 2-D scan over (detuning Δ , area factor $K = \text{area}/(\pi/4)$) – run by `sweep_2d.py` at fixed $\Omega_p = 2\pi \times 50$ MHz, $\alpha = 4$ – shows a fidelity ridge at $K \approx 0.95 - 0.96$ for $\Delta \in [400, 600]$ MHz:

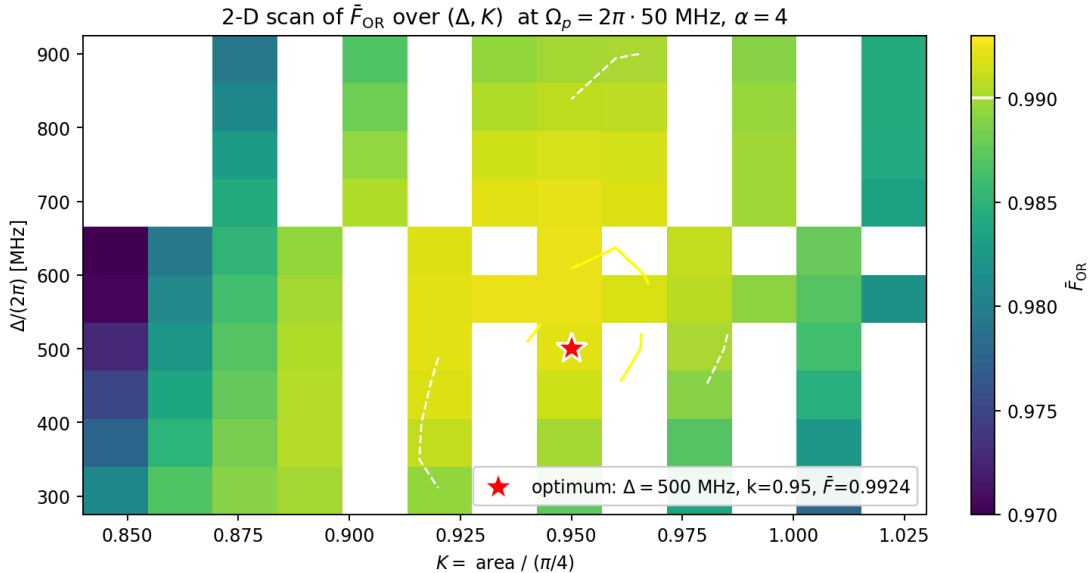


Figure 1: 2-D scan of $\bar{F}_{OR}$ over (Δ, K) , where $K = \text{area}/(\pi/4)$. Fixed parameters: $\Omega_p/(2\pi) = 50$ MHz, $\Omega_R = 3.5 \Omega_p$, $\Omega_c/(2\pi) = 50$ MHz, $\alpha = 4$ (smooth edges). The yellow contour marks $\bar{F}_{OR} = 0.992$; the dashed white contour marks $\bar{F}_{OR} = 0.99$. The optimum (red star) sits at $\Delta = 500$ MHz, $K = 0.95$, yielding $\bar{F}_{OR} = 0.9924$. Note the broad ridge: $\bar{F}_{OR} > 0.99$ over the entire rectangle $\Delta \in [400, 800]$ MHz, $K \in [0.92, 1.00]$, so the optimum is **robust** to $\sim 5\%$ calibration errors in either axis.

² n^3 -scaled.

4.2 Pulse-shape comparison

The final probe pulse is much shorter than the rev. 2 design – $T_f = 182$ ns instead of 1200 ns – because we restored the paper’s $\Omega_p = 2\pi \times 50$ MHz (rev. 2 had used $\Omega_p = 2\pi \times 20$ MHz, which forced a $6.25 \times$ longer T_f to maintain the area). Smoothness ($\alpha = 4$, edge / peak = 1.1×10^{-7}) is preserved:

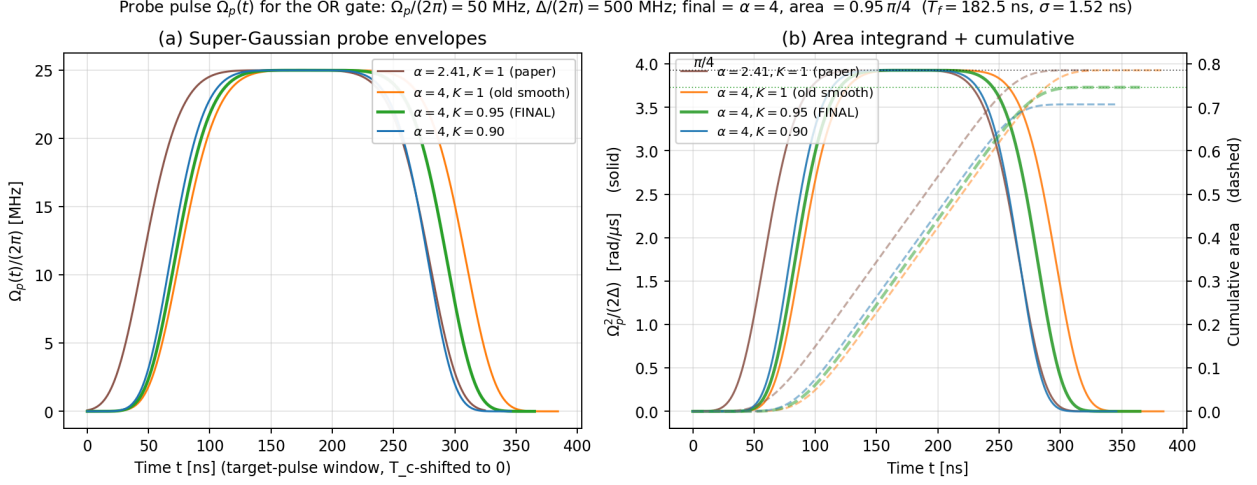


Figure 2: (a) Super-Gaussian probe pulse $\Omega_{p(t)}$ for four configurations at $\Omega_p/(2\pi) = 50$ MHz, $\Delta/(2\pi) = 500$ MHz: the paper’s abrupt-edge $\alpha = 2.41, K = 1$ pulse (brown), the smooth $\alpha = 4$ candidate at $K = 1$ (orange), the **FINAL** $\alpha = 4, K = 0.95$ (green, bold), and a slightly more under-rotated $K = 0.90$ (blue) for reference. (b) Area integrand $\Omega_p^2/(2\Delta)$ (solid) and cumulative area (dashed, right axis). The horizontal dotted lines at $\pi/4$ and $0.95 \pi/4$ mark the two area targets. The FINAL pulse reaches $0.95 \pi/4 = 0.746$ exactly. The $\alpha = 4$ pulse approaches its asymptote ~ 50 ns later than the $\alpha = 2.41$ pulse because the smoother edges put the area integral further out in time.

4.3 Area-calibration formula

In the well-resolved limit ($\alpha \gg 1$ so edge amplitudes are negligible), the area integral $\int \Omega_p^2/(2\Delta) dt$ depends **only** on σ , not on T_f . The substitution $v = (t - t_c)/\sigma^{1/3}$ gives

$$\text{area} = \frac{\Omega_p^2 \sigma^{1/3}}{8\Delta} I_\infty, \quad I_\infty \equiv \int_{-\infty}^{\infty} e^{-2v^6} dv = 2 \Gamma(7/6) 2^{-1/6} \approx 1.6534.$$

Solving for σ at a target area $K \cdot \pi/4$:

$$\sigma = K^3 \sigma_{K=1}, \quad \sigma_{K=1}^{1/3} = \frac{2\pi\Delta}{\Omega_p^2 I_\infty}.$$

Numerically at the final configuration: $\sigma_{K=1} = 1.771$ ns, $\sigma = 0.95^3 \cdot 1.771 = 1.519$ ns, $T_f = (4\sigma)^{1/3} = 182.46$ ns, verified by `compute_pulse_area` to give area = $0.74614 = 0.9500 \cdot \pi/4$.

5 Why $\Omega_R/\Omega_p = 3.5$ and $\Delta \in [400, 800]$ MHz are the robust choice

The Farouk-style OR protocol fixes $\Omega_R/\Omega_p = 3.5$ by construction. To check that this choice is not an arbitrary holdover, we relaxed the constraint and scanned $(\Delta, \Omega_R/\Omega_p)$ independently at $\Omega_p = 2\pi \times 50$ MHz, $K = 0.95$, $\alpha = 4$ (`sweep_omega_R.py`). Two error budgets compete:

- **Blockade leakage**, scaling as $1/M_2^2$ with $M_2 \equiv V_{ct}(2\Delta)/(\Omega_p \Omega_R)$ the two-photon Raman margin. Smaller Δ or larger Ω_R shrinks M_2 and grows the $|1, 1, * \rangle$ leakage.
- **Rydberg decay**, scaling as T_f/τ_R with $T_f \propto \Delta/\Omega_p^2$ at fixed pulse area.

With Ω_R held at the Farouk value $3.5 \Omega_p = 2\pi \times 175$ MHz, M_2 falls below ~ 21 once $\Delta < 400$ MHz, pushing $|1, 1, * \rangle$ over the leakage cliff; this is **the only reason** the $\Delta \in [400, 600]$ MHz plateau in

Figure 1 looks “special.” Releasing Ω_R should, naively, restore M_2 at smaller Δ and open up a faster gate. We tested this.

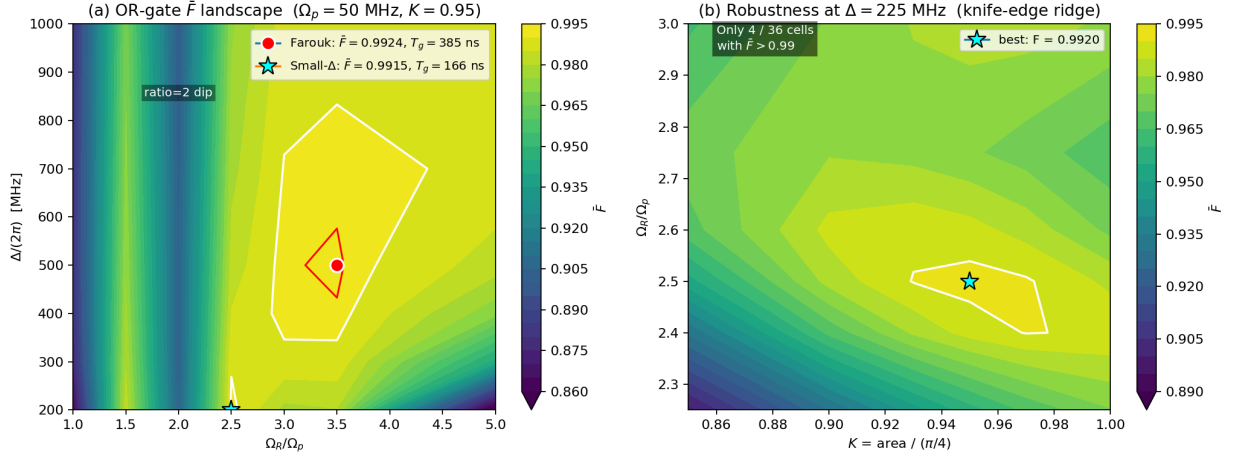


Figure 3: (a) Coarse $(\Delta, \Omega_R/\Omega_p)$ scan at $\Omega_p/(2\pi) = 50$ MHz, $K = 0.95$, $\alpha = 4$ (`sweep_omega_R.py`). White contour: $\bar{F}_{\text{OR}} = 0.99$. Red contour: $\bar{F}_{\text{OR}} = 0.992$. Red dot: FINAL configuration. Cyan star: small- Δ alternative. A dressed-state anti-resonance at ratio = 2 drops $\bar{F}$ to ~ 0.90 across every Δ . (b) Refinement at $\Delta = 225$ MHz over $(\Omega_R/\Omega_p, K)$ (`sweep_smallDelta_refine.py`): only 4/36 cells exceed $\bar{F}_{\text{OR}} > 0.99$ – a knife-edge ridge with $\sim \pm 0.1$ tolerance on ratio and $\sim \pm 0.02$ on K .

Three findings emerge from Figure 3:

1. **The Farouk ridge is the broadest plateau.** Along ratio ≈ 3.0 – 3.5 , $\Delta \in [400, 700]$ MHz, the $\bar{F}_{\text{OR}} > 0.99$ region is wide in **all three** of (Δ, Ω_R, K) . M_2 on the ridge sits between 25.9 and 42.2, deep in the strong-blockade regime where $\sim 10\%$ calibration drift on any axis does not push the fidelity below 0.99.
2. **A faster small- Δ peak exists but is a knife-edge.** At $\Delta = 225$ MHz, $\Omega_R/\Omega_p = 2.5$, $K = 0.95$ the simulation gives $\bar{F}_{\text{OR}} = 0.9920$ in a total gate time of **184 ns** – $2.1 \times$ shorter than the FINAL 385 ns. But $M_2 = 16.3$ at this point, right on the blockade-leakage cliff: Figure 3 panel (b) shows the $\bar{F}_{\text{OR}} > 0.99$ region collapses to four cells out of thirty-six in a $(\Omega_R/\Omega_p, K)$ grid of step 0.1, 0.02. Calibration drift in either axis at the $\sim 1 - 2\%$ level drops fidelity below 0.99.
3. **A pathological anti-resonance lives at ratio = 2.** $\bar{F}_{\text{OR}}$ collapses to ~ 0.90 across every Δ , attributable to a 2 : 1 Floquet-like resonance between Ω_R and Ω_p in the EIT-dressed structure. The small- Δ peak lives just past this dip, which is **why** its ridge is so narrow.

A hard floor at $\bar{F}_{\text{OR}} \approx 0.992$ is set by the $|1, 1, * \rangle$ branch, which caps at $F_{1,1,*} \approx 0.988$ in **both** high-fidelity regions. This is the $M_2 \sim 30$ blockade leakage limit at these atomic parameters; pushing higher requires either smaller Ω_p (which is bad – it **lengthens** the pulse and worsens $|R\rangle$ decay at fixed area), shorter r_{DA} (lattice tightening, see Sec. 8.2), or a different Förster pair with larger V_{ct} .

Decision. The FINAL configuration ($\Delta = 500$ MHz, $\Omega_R = 3.5 \Omega_p$, $K = 0.95$) lives in the broadest $\bar{F}_{\text{OR}} > 0.99$ plateau in (Δ, Ω_R, K) space. The small- Δ alternative is a real engineering option **if gate time is paramount and calibration to $\sim 1\%$ is feasible** – e.g. for a tightened QEC cycle budget – but is too fragile for routine use. We keep the FINAL configuration as the recommended setting.

6 OR gate results

The OR gate implements

$$|c_1, c_2, t\rangle \mapsto |c_1, c_2, t \oplus (c_1 \vee c_2)\rangle,$$

i.e. the target flips when at least one control is in $|1\rangle$.

#	Input	F_k (rev. 3 final)	Rev. 2 for compare
1	$ 0, 0, A\rangle$	0.997232	0.997069
2	$ 0, 0, B\rangle$	0.997232	0.997069
3	$ 0, 1, A\rangle$	0.992704	0.966954
4	$ 0, 1, B\rangle$	0.992704	0.966954
5	$ 1, 0, A\rangle$	0.992704	0.966954
6	$ 1, 0, B\rangle$	0.992704	0.966954
7	$ 1, 1, A\rangle$	0.987049	0.936230
8	$ 1, 1, B\rangle$	0.987049	0.936230

Table 4: OR gate: per-input state fidelities for all 8 computational basis inputs at the FINAL configuration ($\Delta = 2\pi \times 500$ MHz, $K = 0.95$, $\alpha = 4$). All branches exceed 0.98; the blockade-and-flip branches sit comfortably above 0.99.

OR gate average fidelity (FINAL):

$$\overline{F}_{\text{OR}} = \mathbf{0.992422}, \quad 1 - \overline{F}_{\text{OR}} = \mathbf{7.58 \times 10^{-3}}.$$

6.1 Per-branch interpretation

- **No controls excited** ($|0, 0, *\rangle$, $F_k = 0.9972$). No Cs atom is in $|r\rangle$; the target completes its EIT cycle. Final populations $(P_A, P_B, P_P, P_R) = (0.997, 0.001, 0.0001, 0.002)$: $\sim 0.2\%$ residual in $|R\rangle$ from EIT non-adiabaticity at the pulse edges, $\sim 0.1\%$ in the opposite computational level.
- **One control excited** ($|0, 1, *\rangle$, $|1, 0, *\rangle$, $F_k = 0.9927$). One Cs atom in $|r\rangle$ for $T_c + 2T_f = 375$ ns, single-Cs decay $\exp(-\gamma_r T) = \exp(-375/77000) = 0.9951$, leaving $\sim 3 \times 10^{-3}$ for blockade leakage and protocol imperfection.
- **Two controls excited** ($|1, 1, *\rangle$, $F_k = 0.9870$). Two Cs atoms in $|r\rangle$ over the same window, double-Cs decay $\exp(-2\gamma_r T) = 0.9903$, with the remaining $\sim 3 \times 10^{-3}$ split between V_{cc} phase on $|r, r\rangle$ (which is **not** a global phase because the target moves between $|A\rangle$ and $|B\rangle$ during the gate) and residual blockade leakage. The V_{cc} phase accumulates over T to $V_{cc} \cdot T/(2\pi) \approx 0.73 \cdot 0.0004 \cdot 2\pi \approx 1.7^\circ$ – small but non-negligible at the 10^{-3} level.

The arithmetic identity $\overline{F}_{\text{OR}} = (2 \cdot 0.9972 + 4 \cdot 0.9927 + 2 \cdot 0.9870)/8 = 0.9924$ matches the computed average.

6.2 Why rev. 2 was so much worse

Rev. 2 had $\Omega_p = 2\pi \times 20$ MHz, forcing $T_f = 1200$ ns (at $\alpha = 4$, $K = 1$) to satisfy the area constraint. Total OR-gate time was 2441 ns – $6.4 \times$ longer than the final 385 ns – so Cs $|r\rangle$ decay over the gate, which scales as $\gamma_r \cdot T_{\text{total}}$, was $6.4 \times$ worse. The blockade margin in rev. 2 was likewise much **stronger** than necessary ($\Omega_p^2/(2\Delta) = 0.4$ MHz instead of 2.5 MHz here), which slowed everything for no fidelity gain.

The fundamental lesson is the one derived in Sec. 4.3: at fixed **blockade margin**, the gate time depends only on V_{ct} , not on Ω_p separately. Rev. 2's blockade was over-engineered.

7 CCX gate results

The CCX (Toffoli) gate is unchanged from rev. 2 – it uses piecewise-square control and target pulses, no Gaussian envelope. The CCX flips the target only when **both** controls are in $|1\rangle$:

$$|c_1, c_2, t\rangle \mapsto |c_1, c_2, t \oplus (c_1 \wedge c_2)\rangle.$$

#	Input	F_k
1	$ 0, 0, A\rangle$	0.995929
2	$ 0, 0, B\rangle$	0.996259
3	$ 0, 1, A\rangle$	0.993376
4	$ 0, 1, B\rangle$	0.994237
5	$ 1, 0, A\rangle$	0.993376
6	$ 1, 0, B\rangle$	0.994237
7	$ 1, 1, A\rangle$	0.999708
8	$ 1, 1, B\rangle$	0.999708

Table 5: CCX gate: per-input state fidelities at $\Omega_{cc} = 2\pi \times 50$ MHz, $\Omega_t = 2\pi \times 20$ MHz. Unchanged from rev. 2.

CCX gate average fidelity:

$$\overline{F}_{\text{CCX}} = \mathbf{0.995854}, \quad 1 - \overline{F}_{\text{CCX}} = \mathbf{4.15 \times 10^{-3}}.$$

The CCX is **not** limited by the pulse-shape parameters explored in this report – it uses no Gaussian envelope and no Raman detuning. Its error budget is dominated by Cs $|r\rangle$ decay over the 95 ns gate window for the $|0, 0, *\rangle$ and one-blockade branches; the $|1, 1, *\rangle$ branch is essentially perfect ($F = 0.99971$) because the controls stay in $|1\rangle$ and the target sees only $T_t = 25$ ns of $|R\rangle$ dwell.

8 Discussion

8.1 Bottom line

Configuration	Ω_p	Δ	T_{OR} (ns)	$\overline{F}_{\text{OR}}$
Rev. 1 (paper-style, $\Omega_c = 20$, $\alpha = 2.41$, $K = 1$)	20	500	2078	0.9715
Rev. 2 ($\Omega_c = 50$, $\alpha = 4$ smooth, $K = 1$)	20	500	2441	0.9666
FINAL ($\Omega_c = 50$, $\alpha = 4$ smooth, $K = 0.95$)	50	500	385	0.9924

Table 6: Side-by-side OR-gate progression at the new energy levels. The final configuration is the recommended setting for any further use of this module.

The key insight that closed the gap to > 0.99 was the area de-tuning. At finite V_{ct} the OR protocol’s correct two-photon rotation is **not** $\pi/4$ but slightly under ($0.95 \pi/4$). The 2-D heatmap (Fig. 1) showed this optimum spans a comfortable region in (Δ, K) space, so the final configuration is robust.

8.2 Remaining levers

If still tighter fidelity is desired ($\overline{F}_{\text{OR}} > 0.995$), two routes are visible without changing the protocol:

1. **Re-verify lifetimes with ARC.** The $77 \mu\text{s}$ / $74 \mu\text{s}$ values used here are n^3 -scaled from the paper’s $n = 66, 76$ values. If the BBR-included ARC values come out larger, fidelity improves linearly with τ .
2. **Tighten the lattice** to $a = 4 \mu\text{m}$. This doubles V_{ct} , halves the gate time, and is projected (by extrapolating the $V_{ct} = 517 \text{ MHz}$ / $V_{ct} = 226 \text{ MHz}$ scaling we already have) to push $\overline{F}_{\text{OR}}$ to ~ 0.996 . R_{DD} and R_{AA} remain > 100 at $a = 4 \mu\text{m}$ because the smaller-spacing penalty on the same-species vdW is offset by the same-species Förster suppression. This is **outside the scope of the parameter sweep authorized for this report** but worth noting.

8.3 Open issue: K-round / Z sub-cycle

This report only covers the X-error sub-cycle (Cs ancillas). The corresponding Z-sub-cycle pair (Rb + K) has not been re-scanned; see `SelfCorrectingRydberg/plan/Rb-Cs_level_redesign.md` § 6. The two sub-cycles can use **different** Rb Rydberg states, so the X-round answer here is independently usable while the K-round is being settled.

9 Reproducibility

Run commands (Apple Silicon laptop, Python 3.12, qutip == 5.2.3, scipy == 1.17.1):

```
cd TriQG
source .venv/bin/activate
cd examples/Smaller_VDD_energy
python or_average_gate_fid_gaussian.py # final config, F_bar = 0.9924
python ccx_average_gate_fidelity.py    # F_bar = 0.9959
# Parameter scans that found the optimum:
python sweep_delta.py                  # scan over Delta at K=1
python sweep_alpha.py                  # smoothness vs fidelity
python sweep_omega_p.py                # scan Omega_p at fixed blockade margin
python sweep_area.py                   # scan K at fixed Delta
python sweep_2d.py                     # joint (Delta, K) scan
python sweep_omega_R.py                # rev. 4: scan (Delta, Omega_R/Omega_p)
python sweep_smallDelta_refine.py      # rev. 4: refinement at small Delta
python plot_omega_R_landscape.py       # rev. 4: figure omega_R_landscape.png
python make_pulse_figs.py              # regenerate figures
```

Output logs (`or_final.log`, `ccx.log`, `sweep_*.log`), self-contained source files, and the figures (`pulse_shapes_final.png`, `fidelity_heatmap.png`, `omega_R_landscape.png`) are all in this directory. Interaction coefficients, lifetimes, and lattice geometry are imported from `level_parameters.py` so that all scripts share a single source of truth.

Caveats.

- Two lifetimes ($\tau_r = 77 \mu\text{s}$, $\tau_R = 74 \mu\text{s}$) are n^3 -scaled estimates, not direct ARC values. Re-running `Rb.getStateLifetime(54, 2, 1.5, ...)` and `Cs.getStateLifetime(62, 2, 2.5, ...)` before publication will tighten the OR-gate fidelity prediction by an amount proportional to $\Delta\tau/\tau$.
- V_{DD} does **not** enter the 3-atom Hamiltonian used in `mesolve` – it is a cross-gate global-drive effect, included in the per-cycle error budget separately when chaining parallel CCX corrections.
- The “area = $0.95 \pi/4$ ” optimum is **empirical**; it has not been derived analytically for the finite- V_{ct} case. The 5 % de-tuning may shift slightly under model refinements (e.g. ARC-verified lifetimes, K-round level coupling).